Building Client Apps with .NET 6

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David Ortinau

@davidortinau

24 yrs. web, interactive, mobile

Xamarin MVP 2013 - 2017

BA English, Maryville University

Principal Program Manager
~~Visual Studio Mobile Developer Tools~~
~~Xamarin~~
.NET MAUI

Microsoft

.NET MAUI PM Team

- David Ortinau
  david.ortinau@microsoft.com
  @davidortinau
  .NET MAUI SDKs, Xamarin SDKs
- Dmitry Lyalin

  @LyalinDotCom
  XAML, Hot Reload
- Maddy Leger

  @maddyleger1
  .NET MAUI Tooling, Hot Restart
- Jake Kirsch

  @ThisIsKirsch
  Single Project, Getting Started
- Chris Hardy

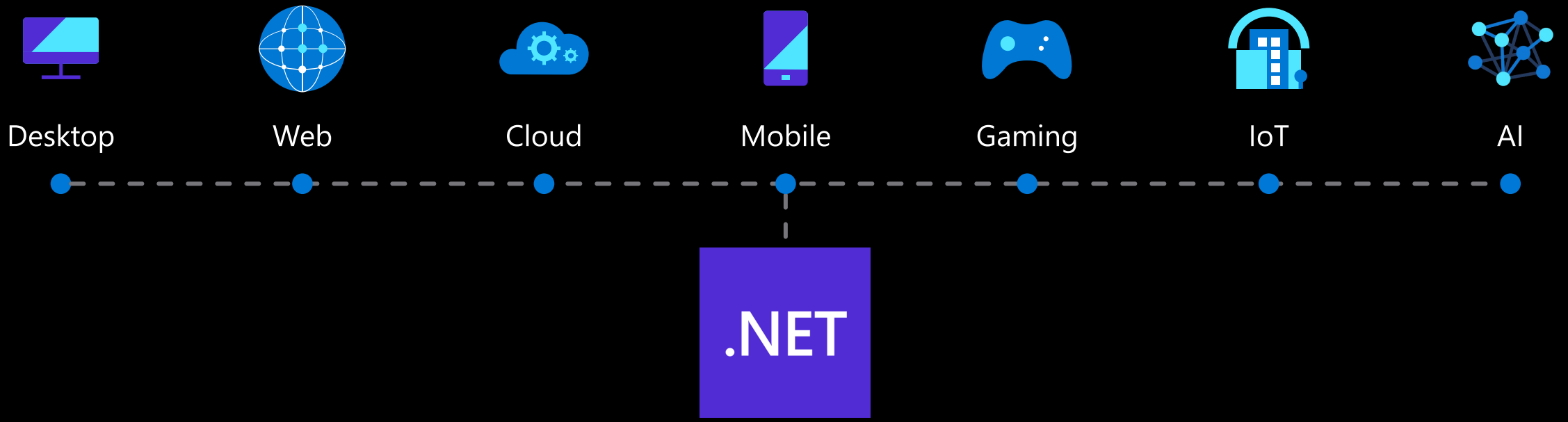
  @chrisntr
  Client App - Manager
- Olia Gavrish

  @oliagavrysh
  Client App Tooling: WPF and WinForms
- Zeren Aksu

  @BilgeZerenAksu
  XAML Experiences

.NET

Your platform for building anything



.NET ecosystem momentum

>5M

.NET developers

In Visual Studio family

#1

Most Loved Framework (2019 & 2020)

.NET Core

Top 30

Highest velocity OSS project

github.com/dotnet

Top 5

Language on GitHub

C#

>10X

Faster than Node.js

ASP.NET Core

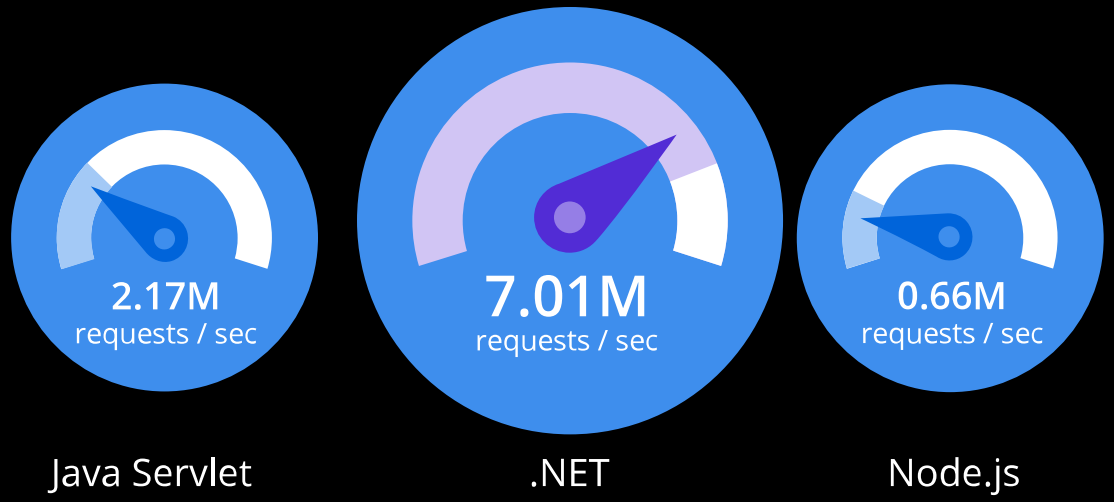
40%

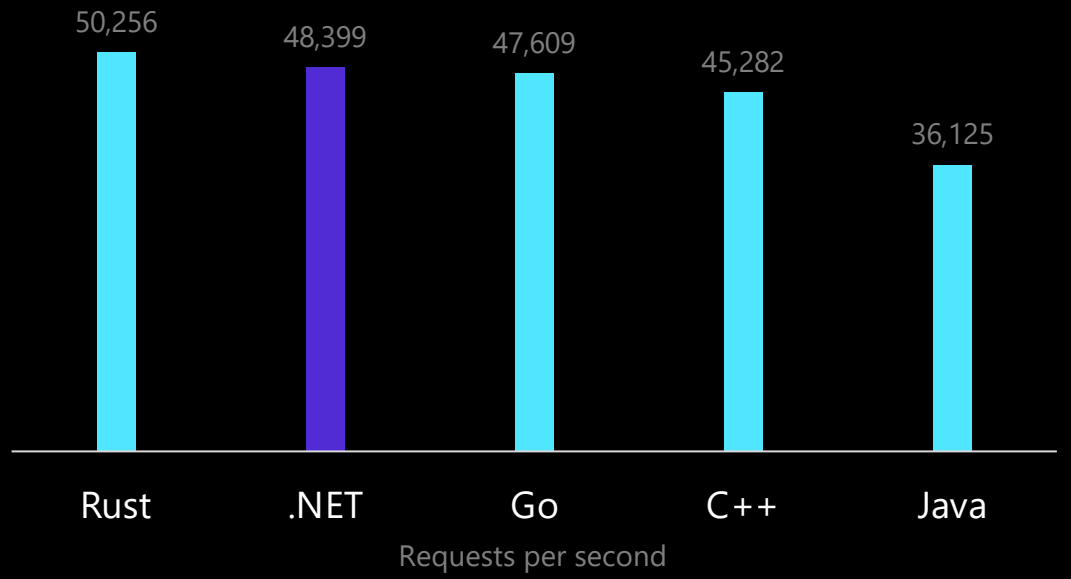
New to .NET are students

dot.net download survey

Sources: insights.stackoverflow.com/survey/2019#most-loved-dreaded-and-wanted; insights.stackoverflow.com/survey/2020#most-loved-dreaded-and-wanted; github.com/cncf/velocity/blob/master/reports/top_projects_by_activity.txt; octoverse.github.com/; www.techempower.com/benchmarks/#section=data-r20&hw=ph&test=plaintext

.NET 5 Performance





>10X faster than Node.js

ASP.NET Core web framework

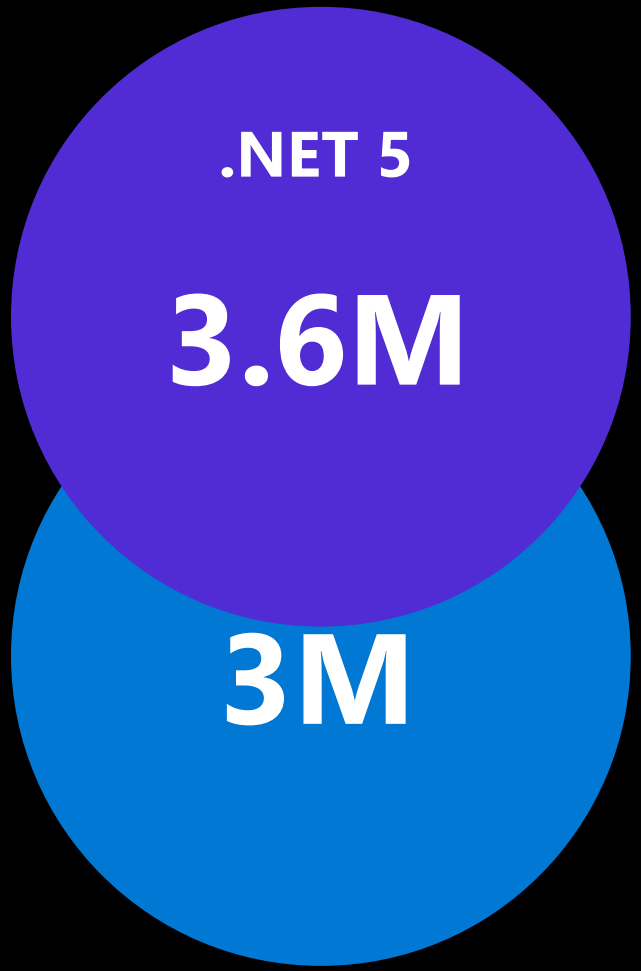
Faster than Go, C++ and Java

gRPC server performance

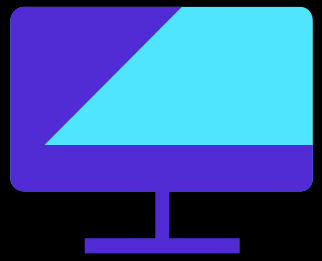
Sources:
https://www.techempower.com/benchmarks/#section=data-r20&hw=ph&test=plaintext
https://github.com/LesnyRumcajs/grpc_bench/wiki/2020-08-11-bench-results

.NET adoption

Downloads in first 6 months of release

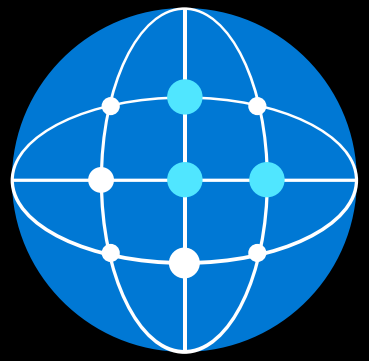


What I mean by “Client App”

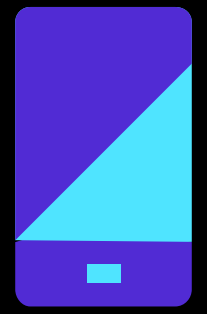


Windows

macOS



All the Browsers



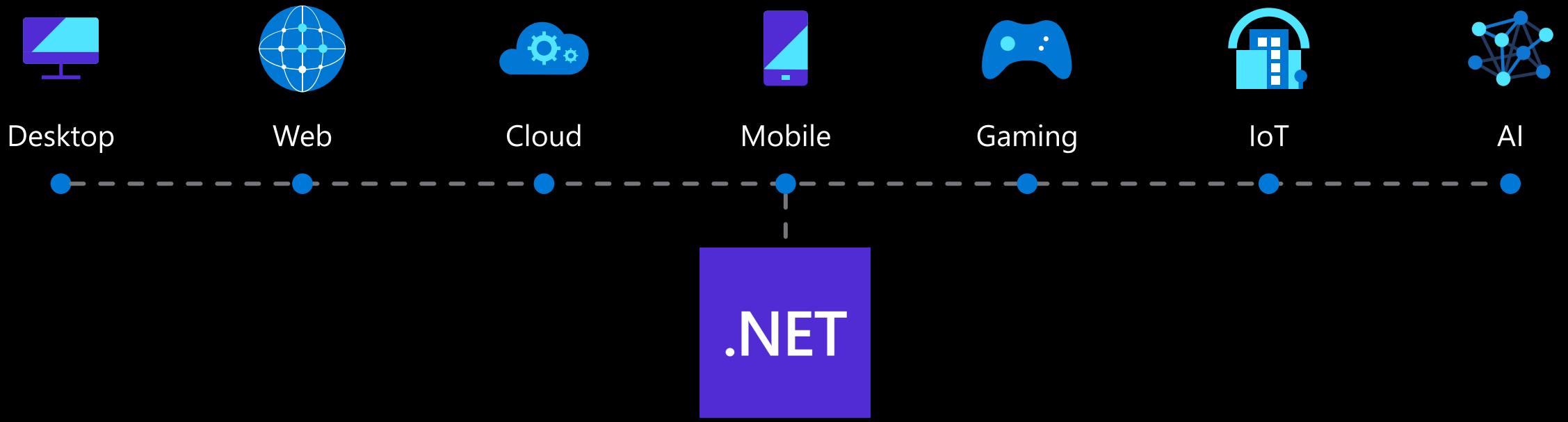
Android

iOS

How do you choose the best way
to build your next project?

.NET

Your platform for building anything



By Reach

	Windows	macOS	Browser	Android	iOS
.NET MAUI	✓	✓		✓	✓
.NET MAUI Blazor	✓	✓	✓	✓	✓
Blazor			✓		
ASP.NET			✓		
Windows App SDK	✓				
WPF	✓				
WinForms	✓				

* Samsung Tizen is also supported with .NET MAUI

By Skill

	C#	XAML	Razor	HTML	JS	Hot Reload	Designer
.NET MAUI	✓	✓		✓	✓	✓	
.NET MAUI Blazor	✓	✓	✓	✓	✓	✓	
Blazor	✓		✓	✓	✓	✓	
ASP.NET	✓		✓	✓	✓	✓	
Windows App SDK	✓	✓				✓	
WPF	✓	✓				✓	✓
WinForms	✓					✓	✓

✓ - optional
✓ - primary

By Other Criteria

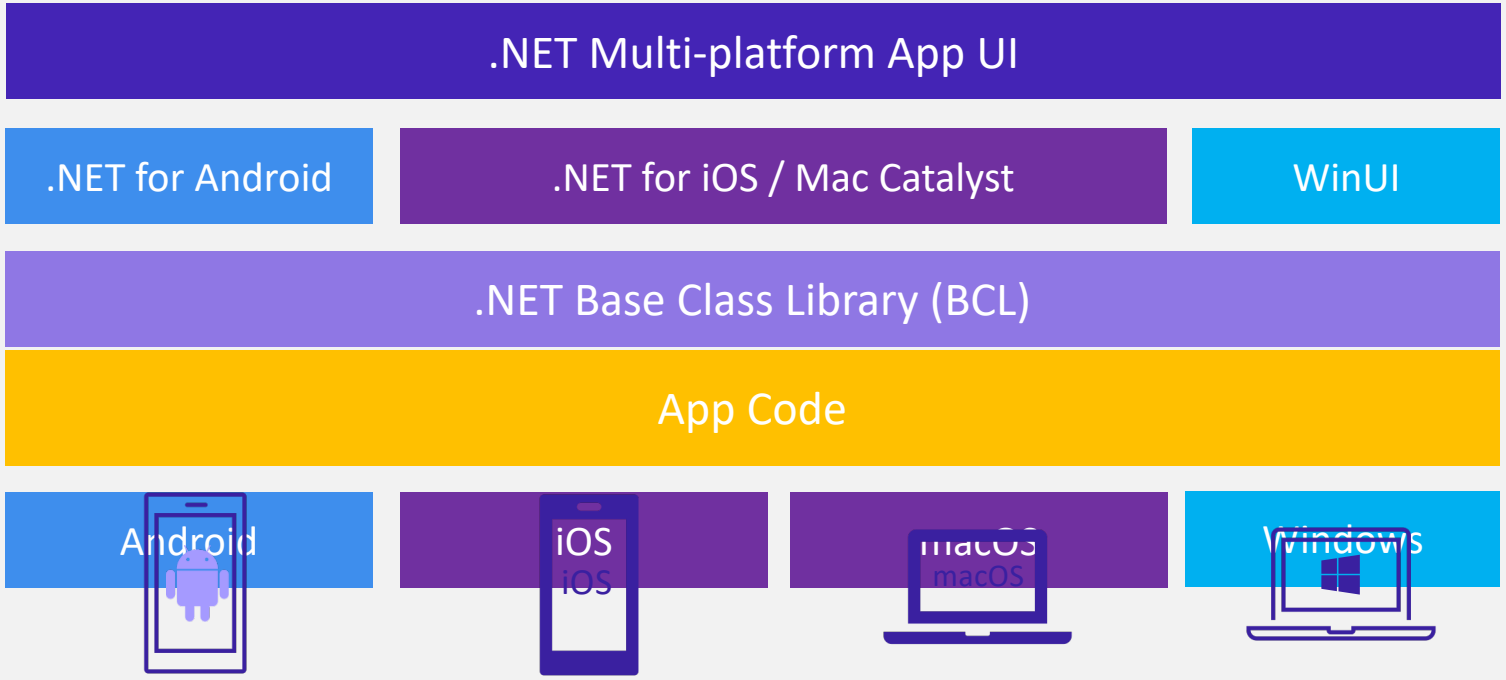
- Testability
- Popularity / Community
- Educational resources
- Quality
- Performance
- Career path
- _____

Let’s simplify

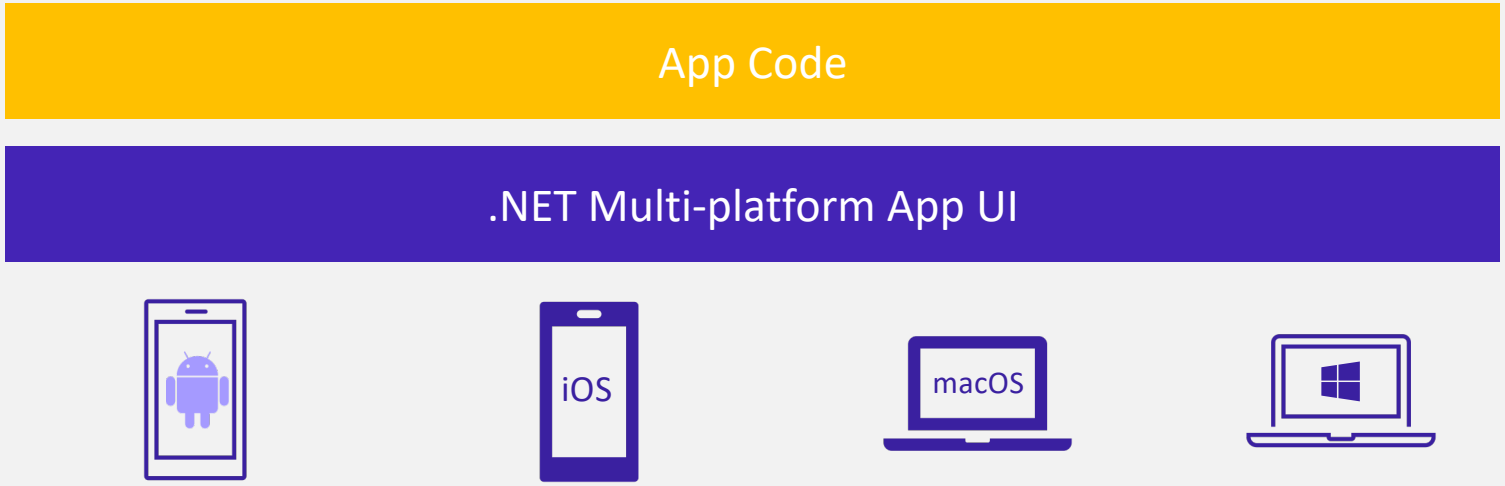
.NET MAUI is the most productive way to develop native apps that perform great on any device that runs Android, iOS, macOS, or Windows from a single codebase.


Weather TwentyOne
Wind Map
78%
From
Pond Elementary
0.2in
From
Pond Elementary
2%
From
Pond
78%
From
City H
5:19
Redmond, WA
52°F
Clear
Next 24 Hours
11 PM 12 AM 1 AM 2 AM 3 AM 4 AM 5 AM
47° 47° 48° 49° 52° 53° 58°
Daily Forecasts
iPhone 12
iOS 14.5
3:58
Redmond, WA
Favorites
18°
Redmond, WA
USA
28°
St. Louis, MO
USA
WeatherTwentyOne
Redmond, WA
52°F
Clear
Winds
14|25
78%
From
Pond Elementary
0.2in
From
Pond Elementary
2%
From
Pond Elementary
9mph
From
Pond Elementary
78%
From
City Hall
0.2in
From
Rockwood
Reservation
2%
From
County Library
Next 24 Hours
9 AM 10 AM 11 AM 12 PM 1 PM 2 PM 3 PM 4 PM 5 PM 6 PM 7 PM
47° 47° 48° 49° 52° 53° 58° 63° 64° 65° 68°
Daily Forecasts
Thu Fri Sat Sun Mon Tue Wed Thu Fri Sat Sun
77° 82° 77° 80° 61° 68° 67° 77° 82° 77° 80°
52° 61° 62° 57° 49° 49° 47° 52° 61° 62° 57°

How does .NET MAUI work?

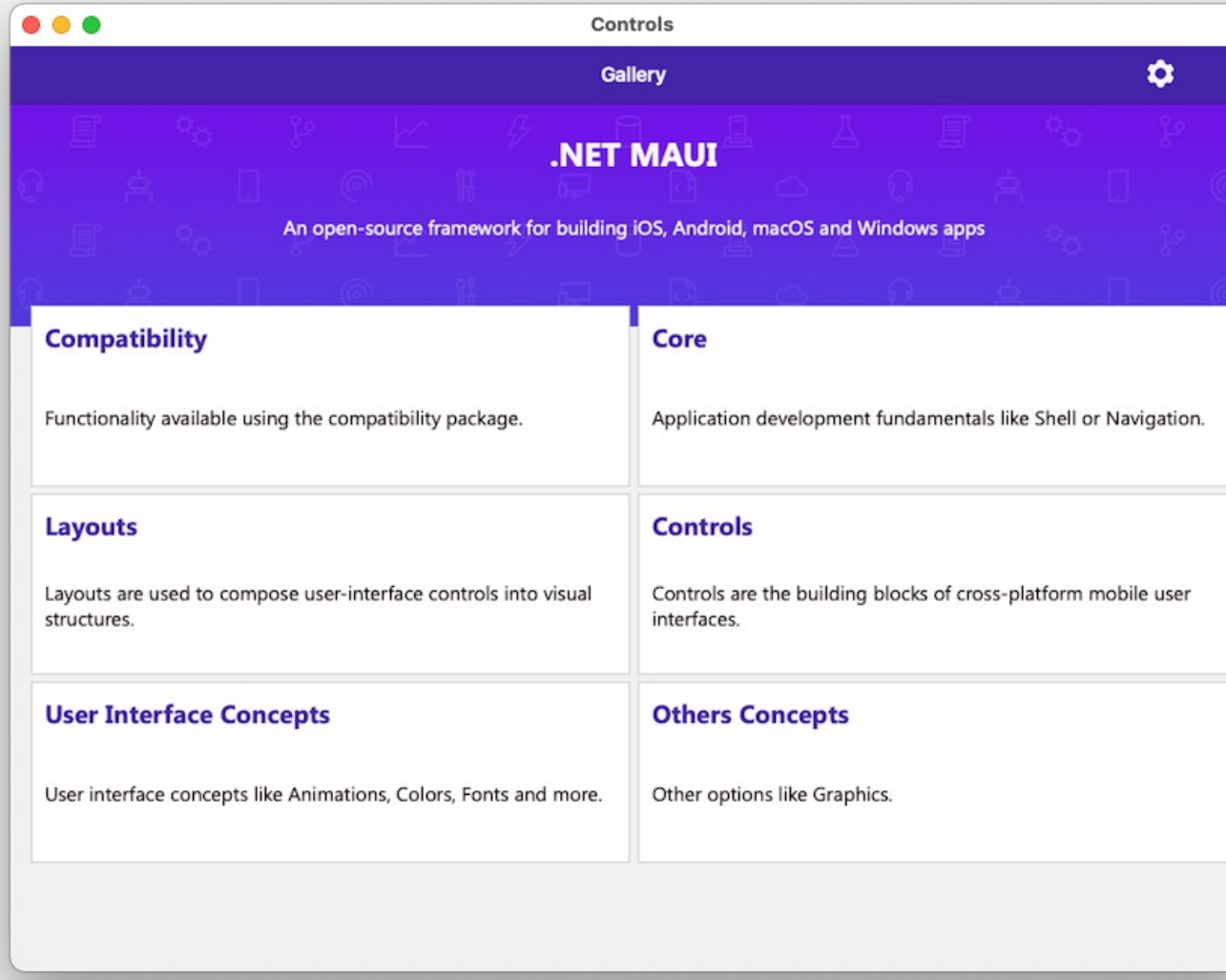


How does .NET MAUI work?

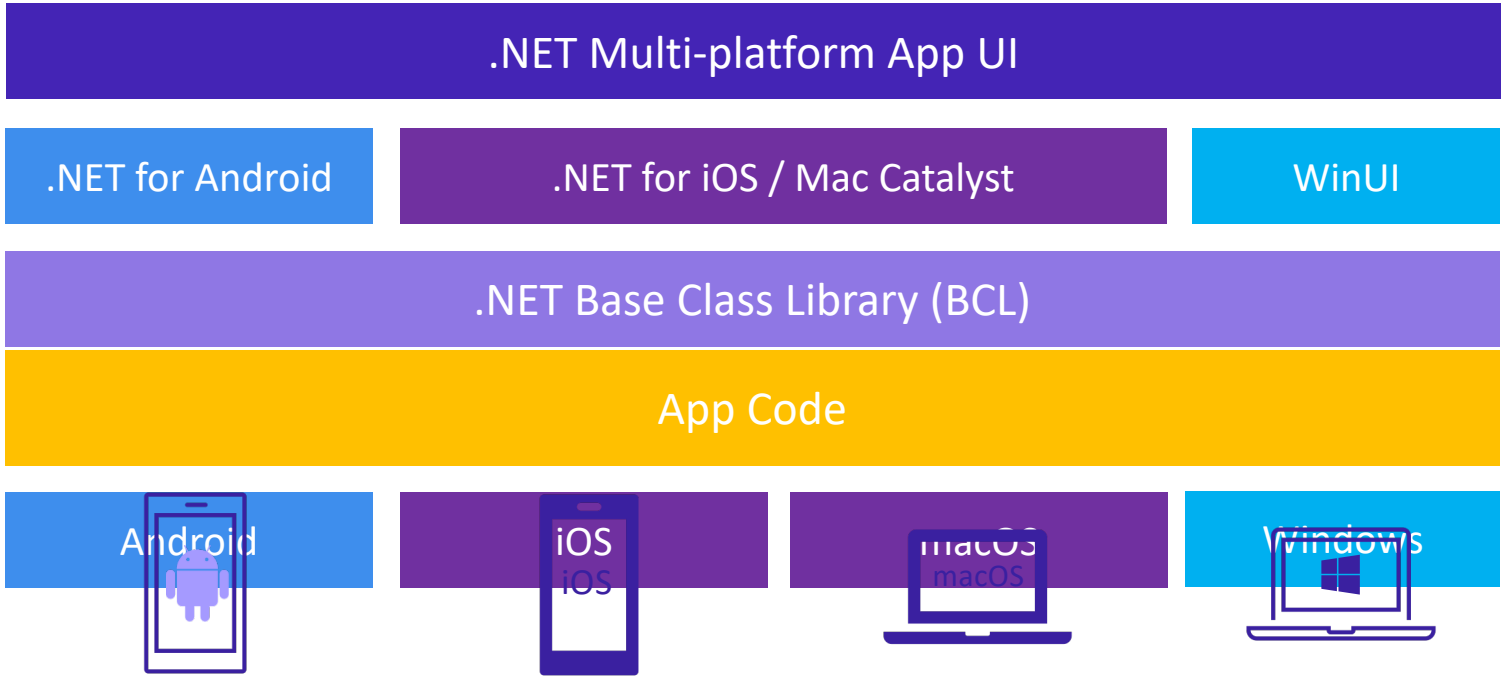


Rich Library of Controls

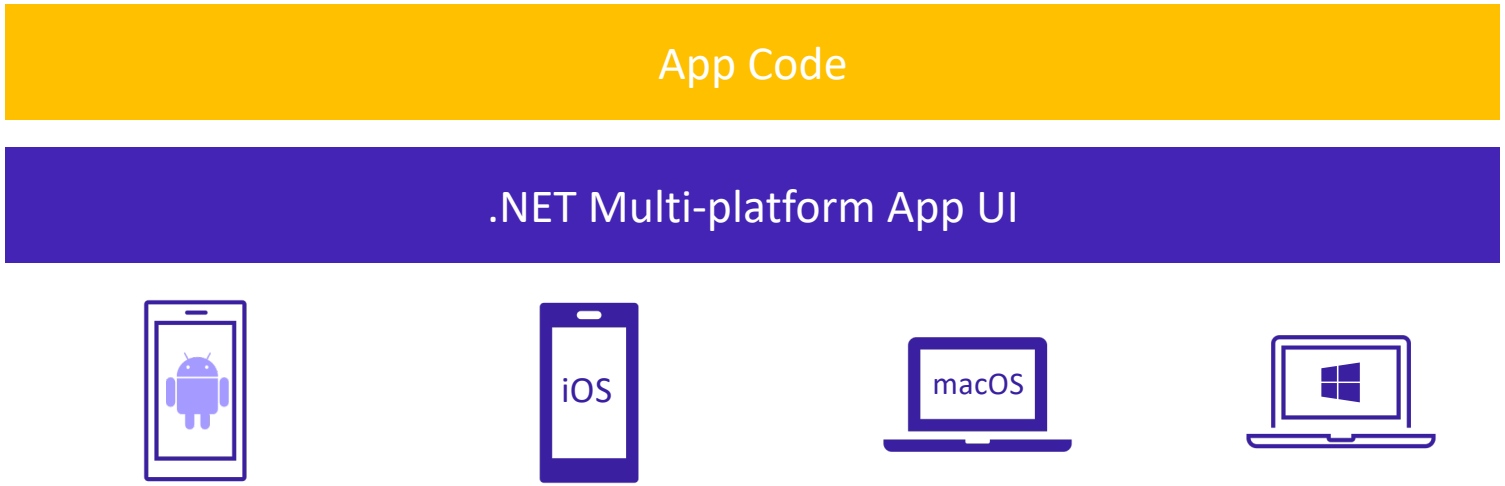
- 40+ Controls
- Navigation
  - Flyout Menu
  - Bottom Tabs
  - Top Tabs
  - Hierarchical
- Adaptive layouts
- Right-to-Left
- Semantic Accessibility



How does .NET MAUI work?

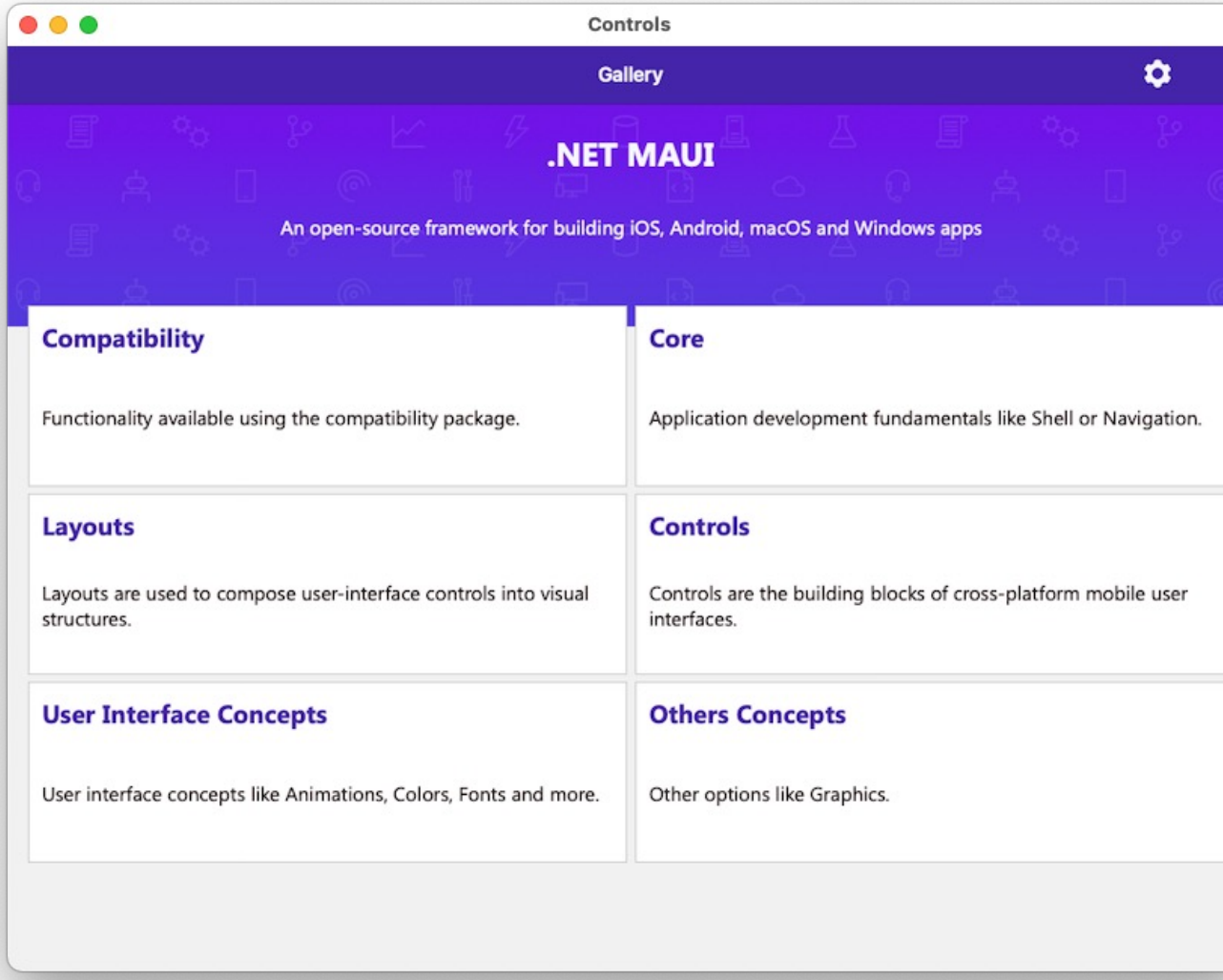


How does .NET MAUI work?



Rich Library of Controls

- 40+ Controls
- Navigation
  - Flyout Menu
  - Bottom Tabs
  - Top Tabs
  - Hierarchical
- Adaptive layouts
- Right-to-Left
- Semantic Accessibility



Component Vendors



















Building UI in .NET MAUI

Optimized for XAML, C#, and MVVM

```xaml
<StackLayout>
    <Label Text="Welcome to Maui!" />

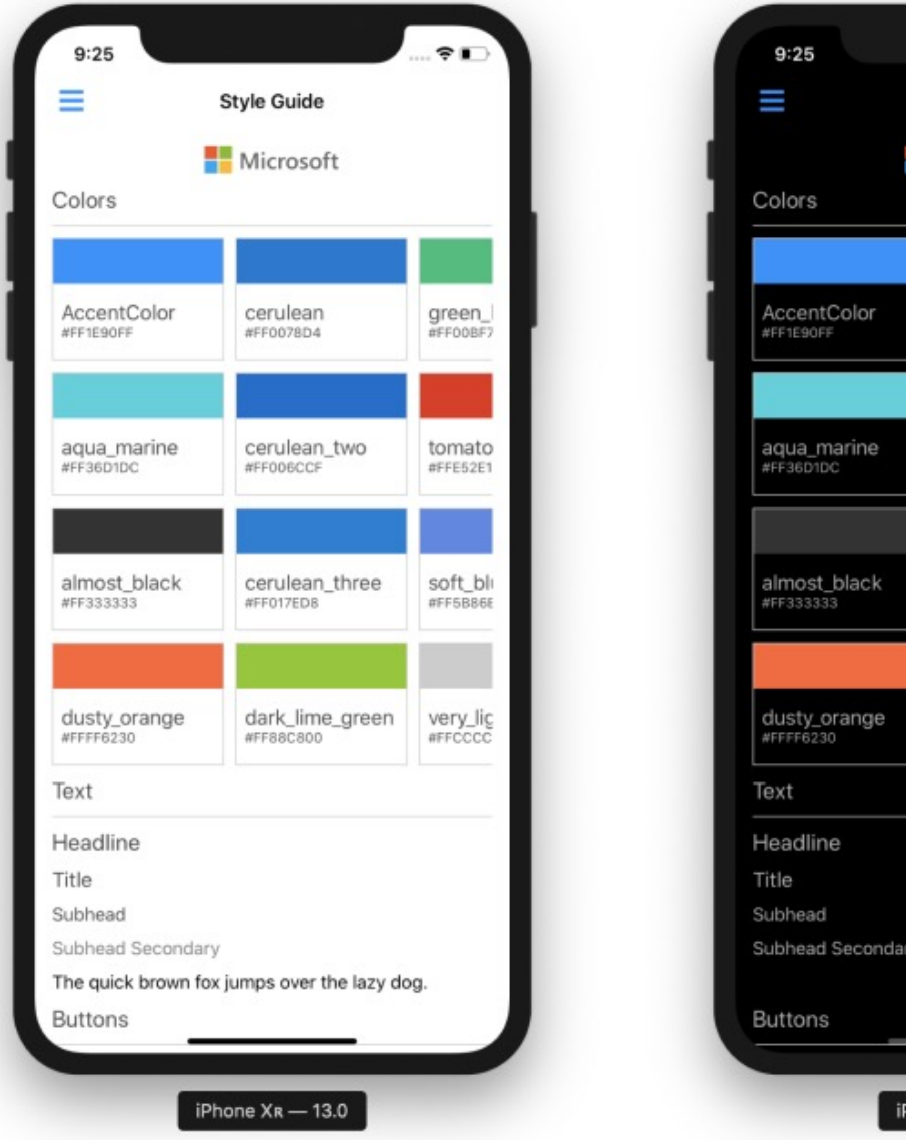
    <Button Text="{Binding Text}"
            Clicked="{Binding ClickCommand}" />
</StackLayout>
```

```
0 references
public ICommand ClickCommand { get; }
1 reference
public string Text { get; set; } = "Click me";

int count = 0;
0 references
void ExecuteClickCommand()
{
    count++;
    Text = $"You clicked {count} times.";
}
```

Abundant Styling Options

- Themes for dark and light
- Static and dynamic styles
- XAML and CSS
- Visual State Manager
- Custom fonts and font icons
- Images from many formats including SVG



Easily customize controls

```
#if ANDROID
Handlers.EntryHandler
    .EntryMapper[nameof(IEntry.BackgroundColor)] = (h, v) =>
    {
        (h.NativeView as global::Android.Views.Entry).UnderlineVisible = false
    };
#endif
```

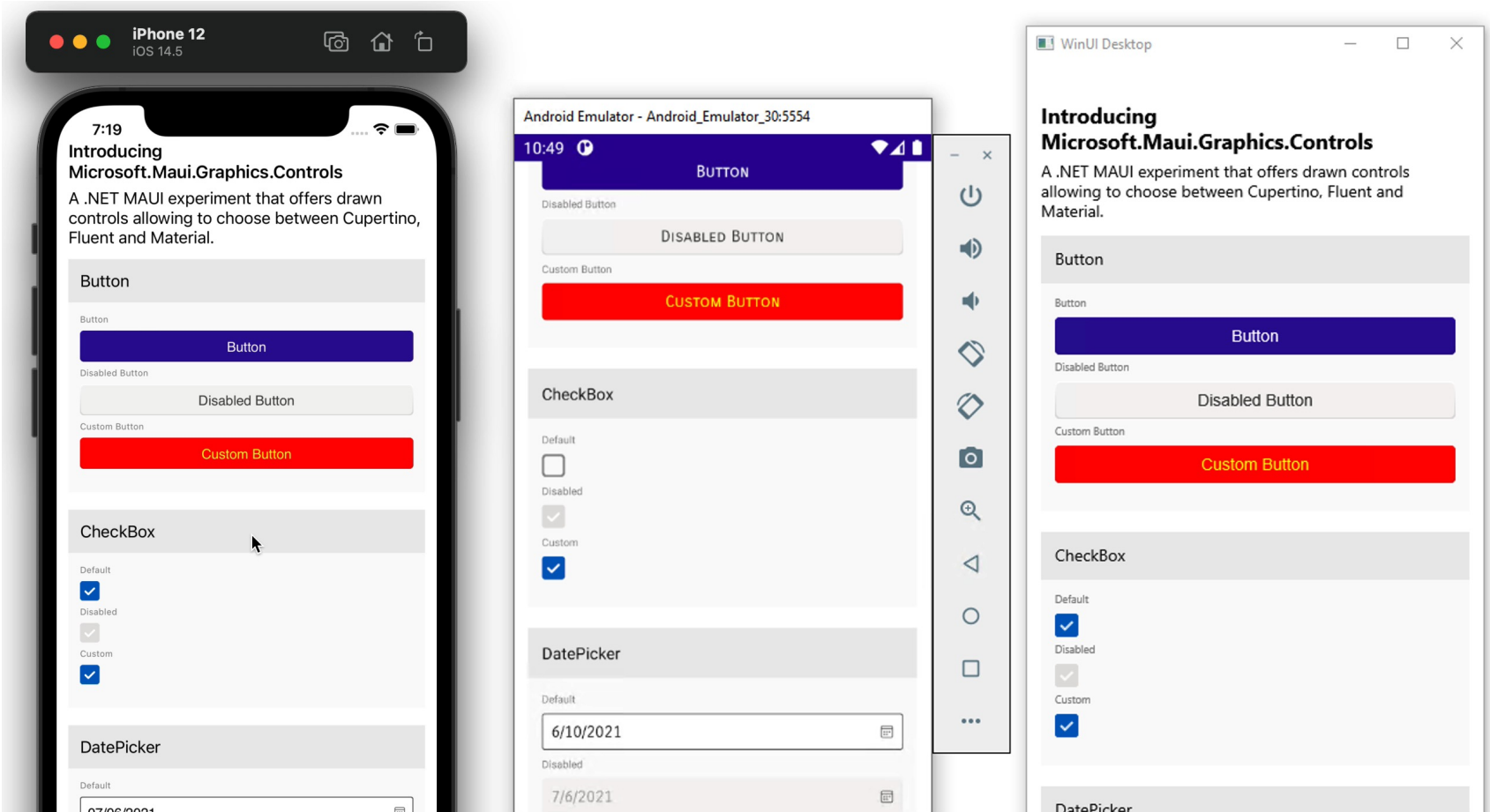
#ifdef if platform specific

Access the handler from anywhere

Dictionary of all props

Drawn UI with Maui.Graphics (Experimental)

- Native graphics or SkiaSharp
- Fluent and Material design



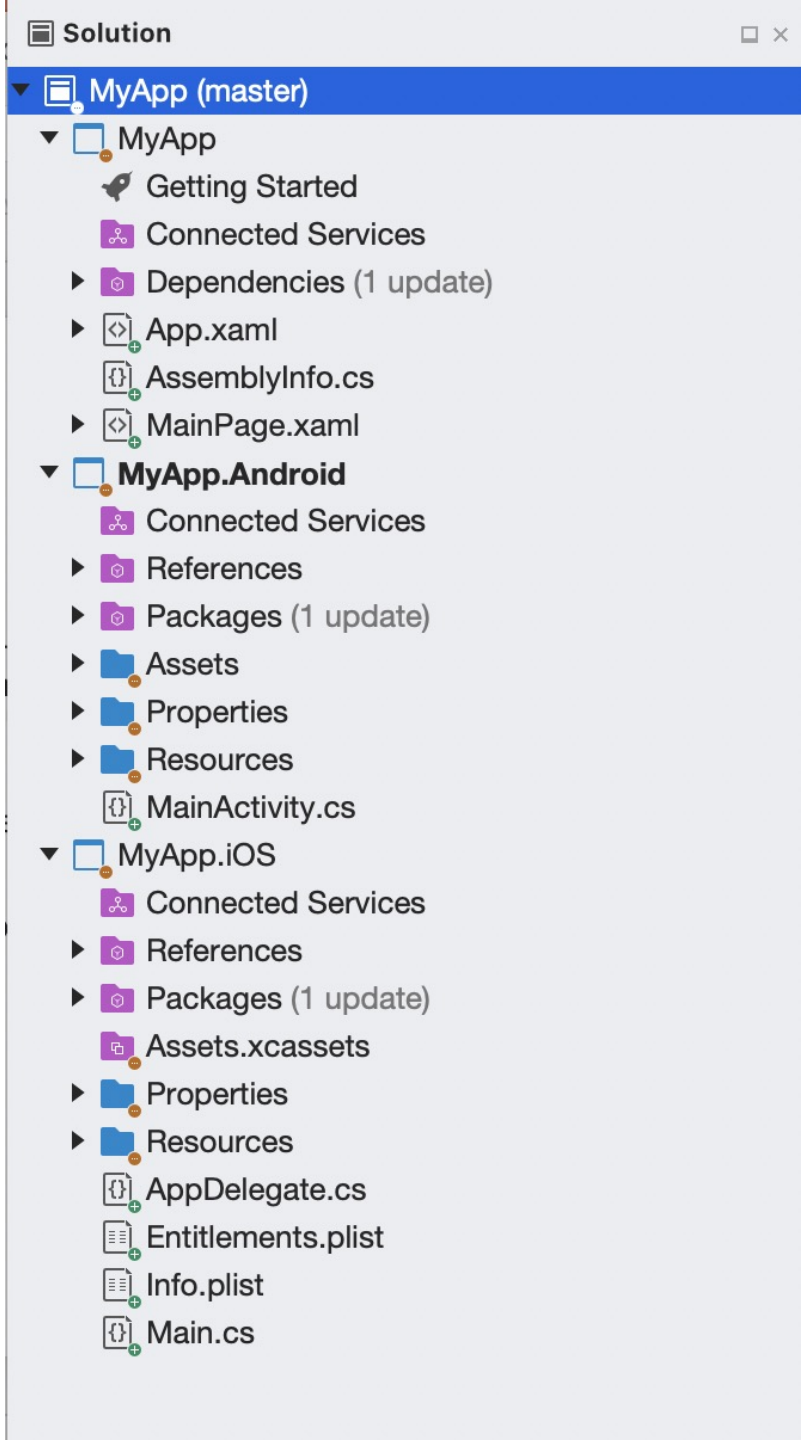
Project System Improvements

Optimized for Cross-Platform Workflows

Single Project

Reducing multiple projects to a single project alleviates several points of friction when developing apps that target multiple platforms.

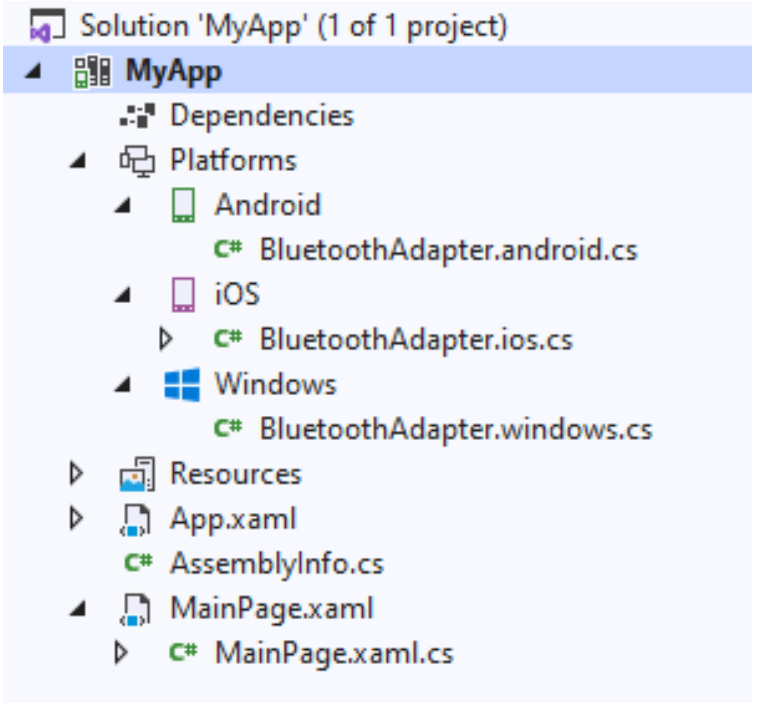
- Images
- Fonts
- Platform code
- Splash screen



Single Project

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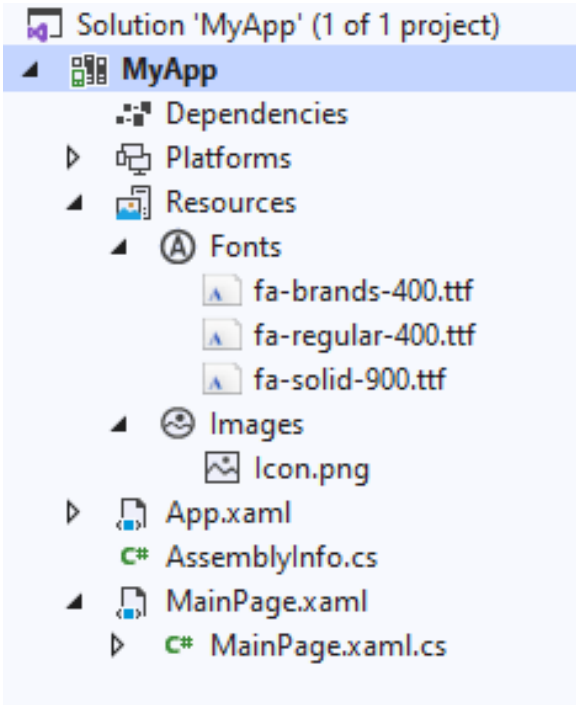
- Images
- Fonts
- Platform code
- Splash screen



Single Project

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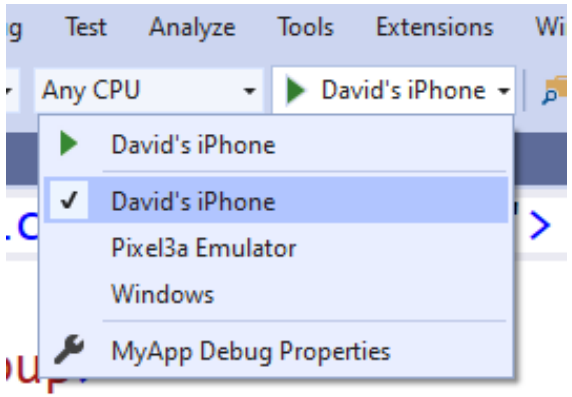
- Images
- Fonts
- Platform code
- Splash screen



Single Project

Reducing multiple projects to a single project alleviates several points of friction when developing apps that target multiple platforms.

- Images
- Fonts
- Platform code
- Splash screen



Faster Development

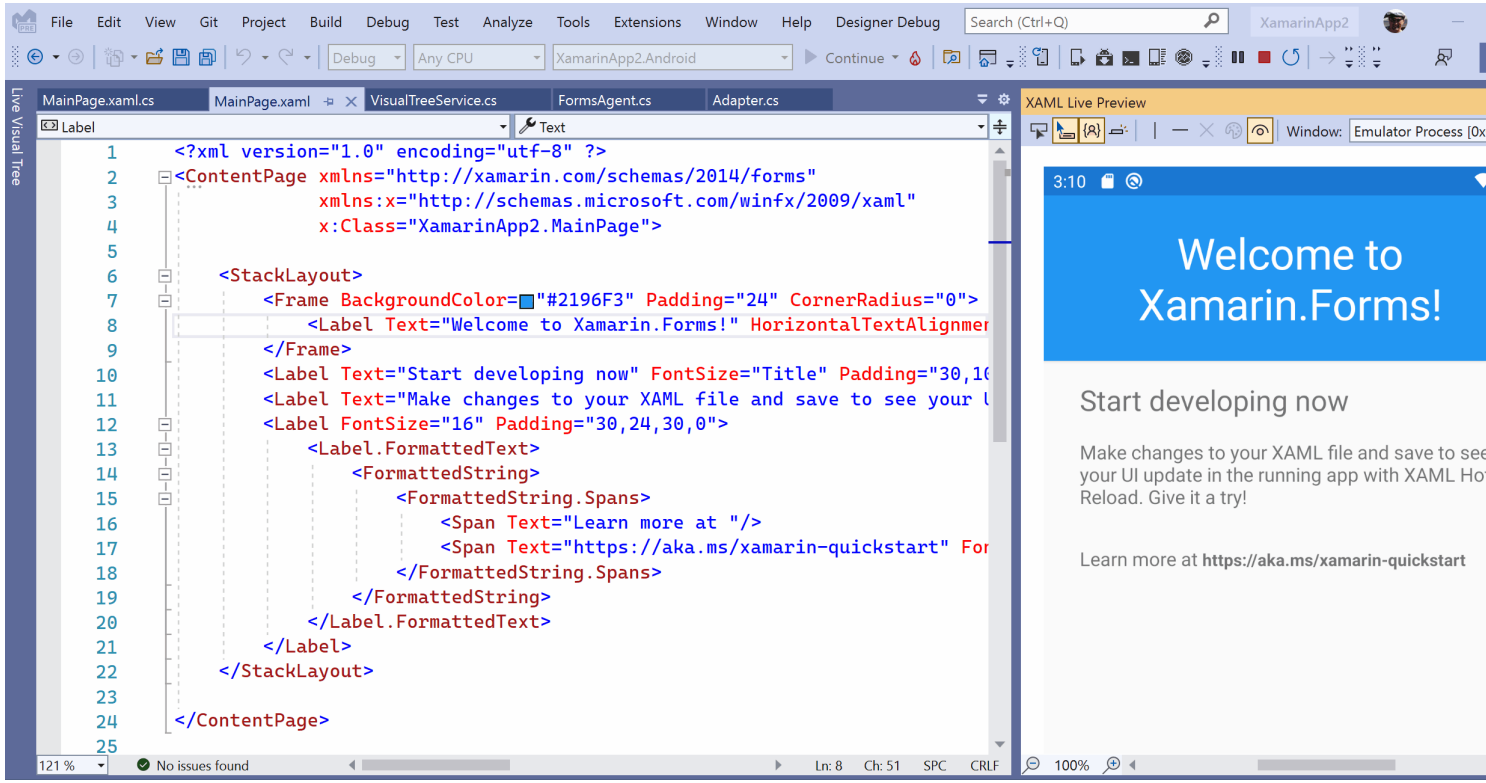
Optimized Inner-loop

Code it Live

Whether XAML or C#, you can code while debugging your application and see the changes instantly. No more stop-edit-rebuild for simple tweaks.

When you do something more complex, Visual Studio will prompt you to rebuild.

- C#
- CSS
- Razor
- XAML



Windows and iOS

Connect any iOS device directly to Windows and start coding in Visual Studio 2022. That's it!

Changes are pushed from Visual Studio directly to the app running on device, no Mac build host required.

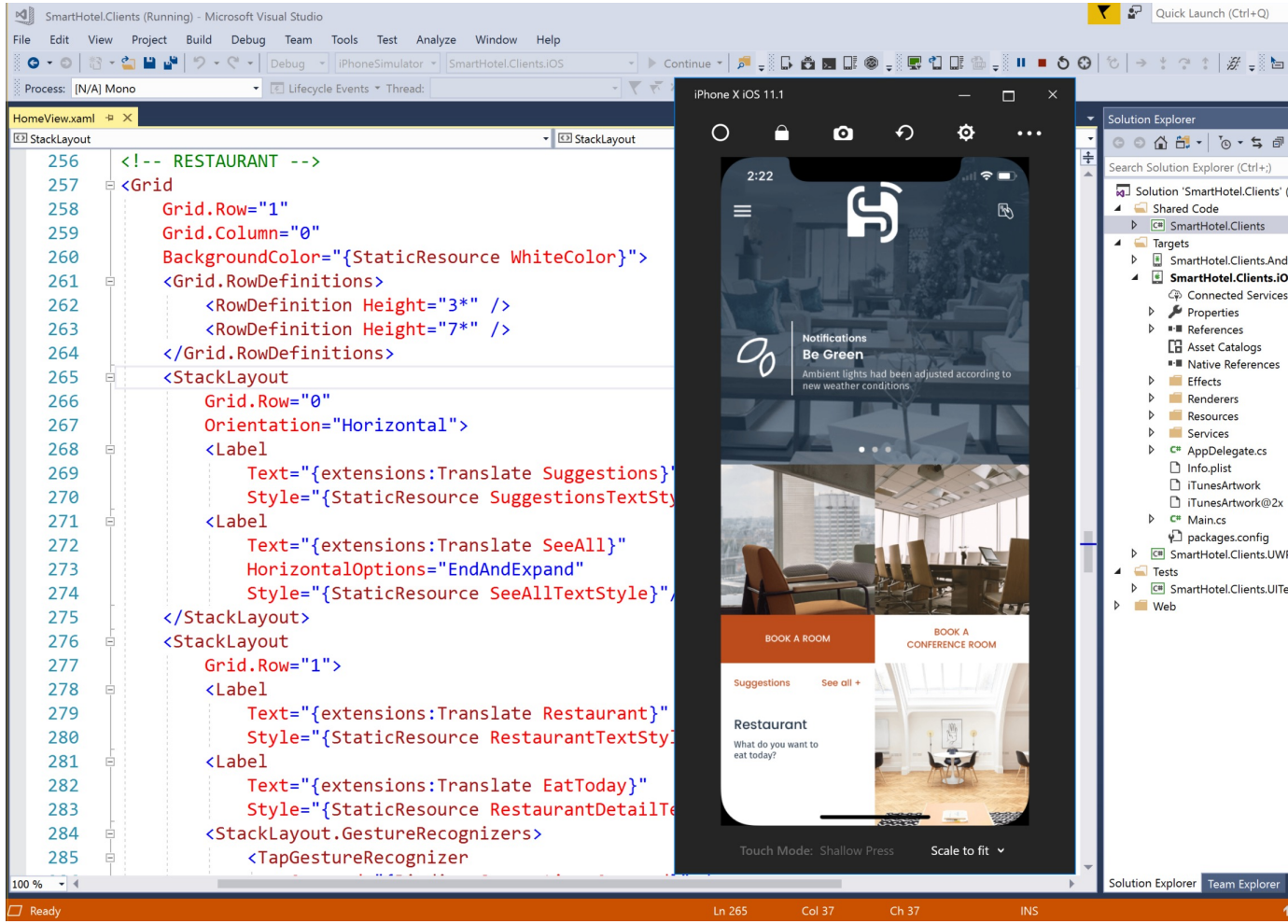
When you're ready to publish and distribute, connect a Mac or use a cloud service to finish the job.



Remote iOS Simulator

For more Apple features, connect your Visual Studio 2022 on Windows to a Mac on your local network. With this you get:

- Native Xcode builds over SSH
- Remote simulator
- Touchscreen support



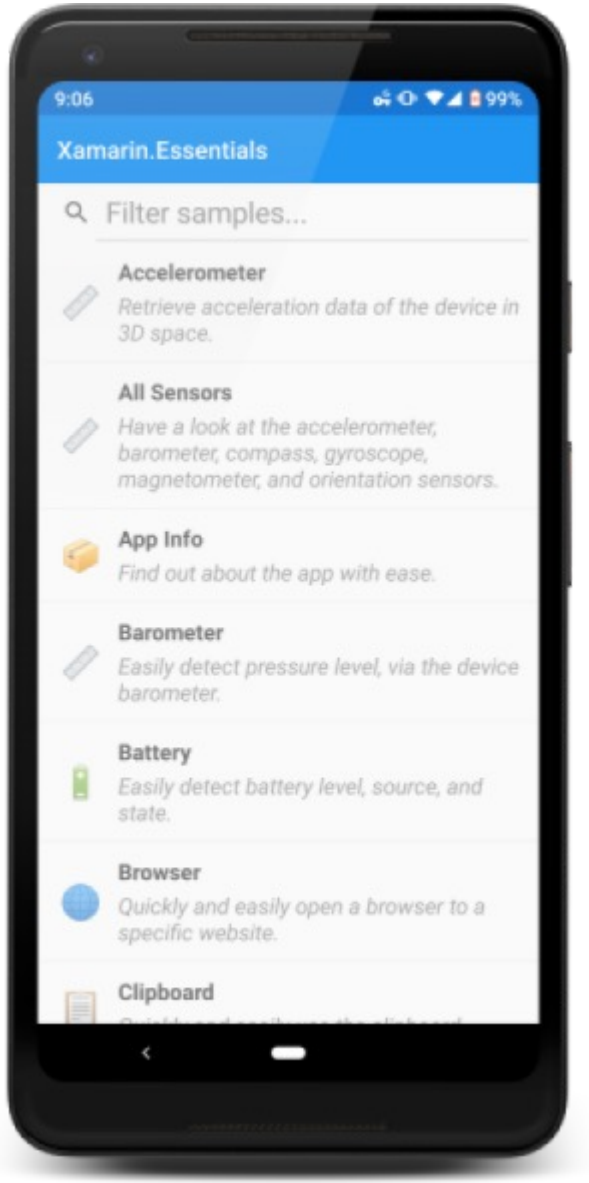
And beyond

More ways to use .NET MAUI

Essentials

Use Maui.Essentials to access device services across all .NET MAUI targets.

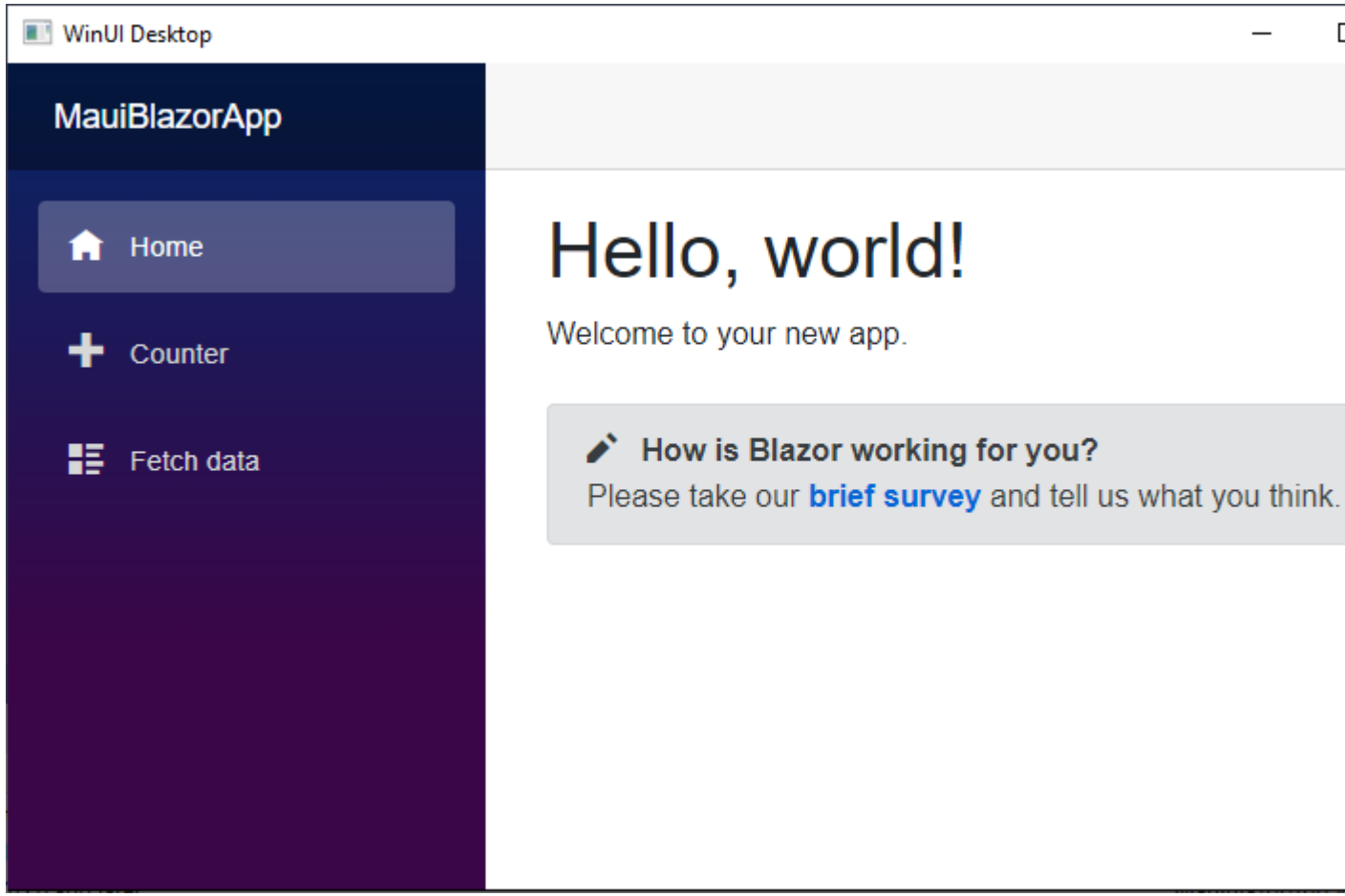
- Next version of Xamarin.Essentials
- Accelerometer
- App Actions
- App Information
- Barometer
- Batter
- Clipboard
- Compass
- Connectivity
- and on and on



Blazor Hybrid

Mix your Blazor app and .NET MAUI to create seamless hybrid experience on desktop and mobile.

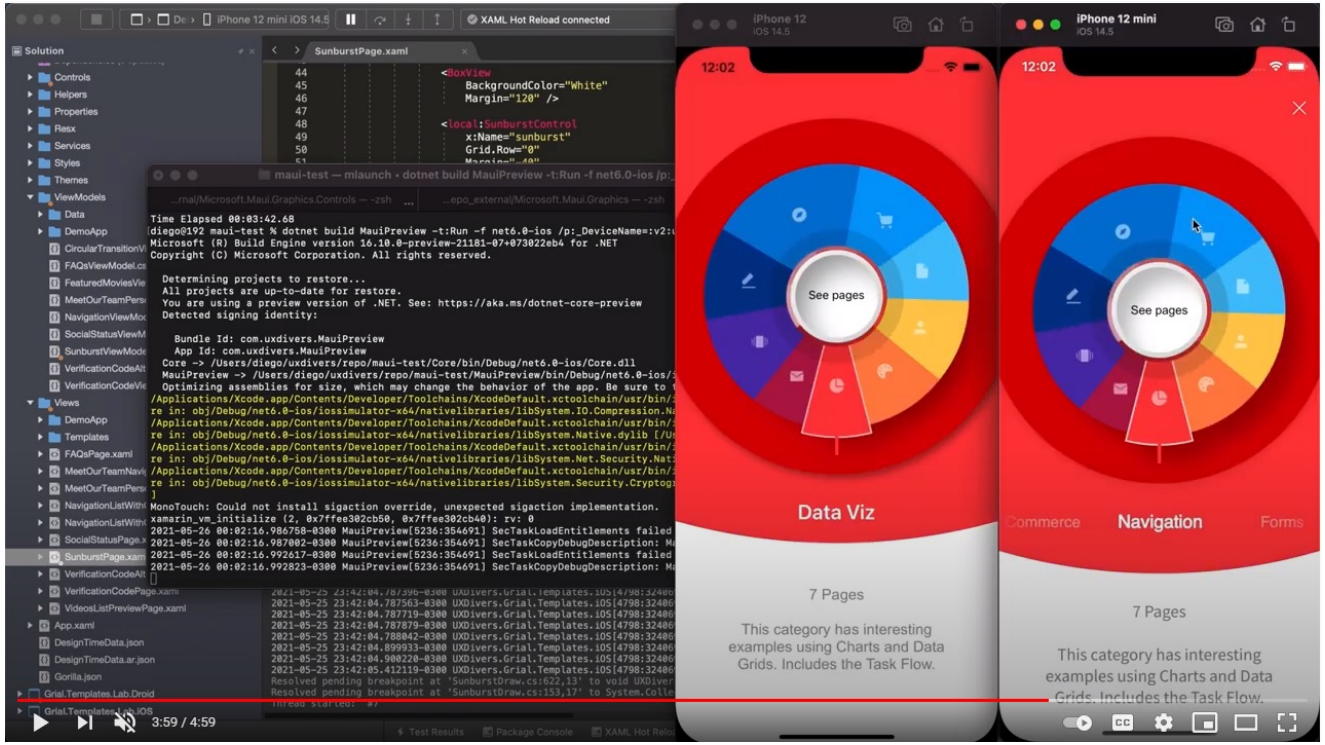
- BlazorWebView hosts web app
- Access any .NET MAUI api from Razor
- Combine web UI and native UI
- Low memory footprint



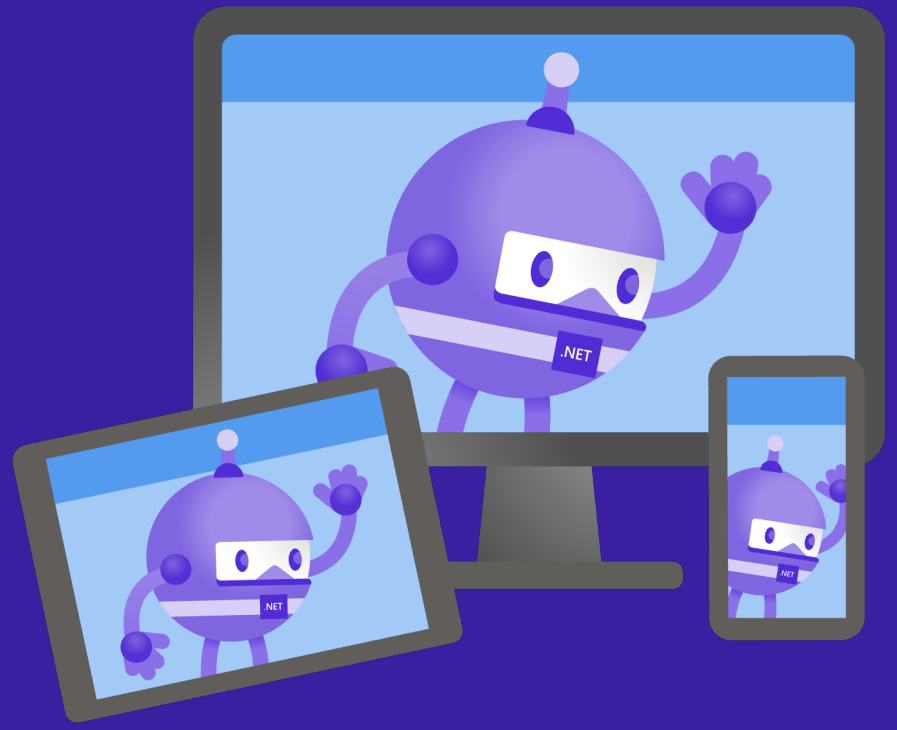
Microsoft.Maui.Graphics

What .NET MAUI is to platform UI toolkits, Maui.Graphics is to platform graphics rendering.

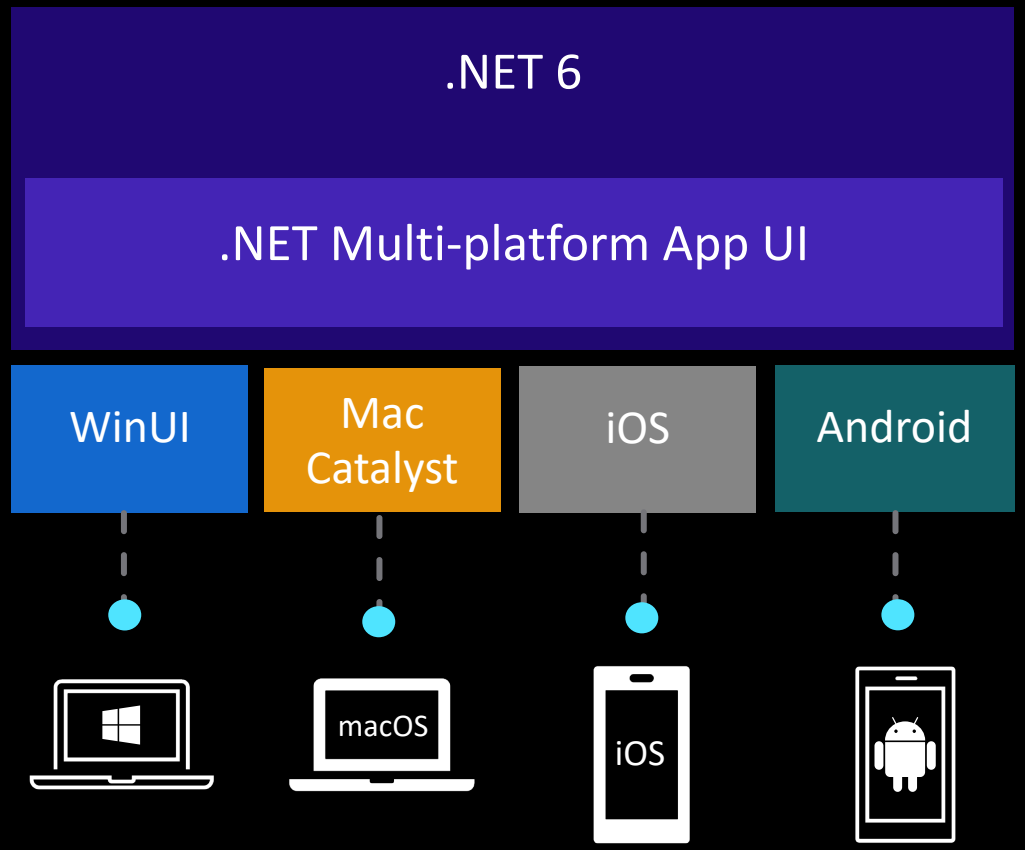
- Canvas with primitive drawing types
- Fonts
- Attributed text
- Bitmaps
- PDFs
- Unified Color types
- Broad platform support
- SkiaSharp supported



Let’s Demo



.NET Multi-platform App UI



Cross-platform, native UI

Single project system, single codebase

Deploy to multiple devices, mobile & desktop

General availability in .NET 6

github.com/dotnet/maui

Visual Studio 2022

- 64-bit
- Faster
- .NET Hot Reload
- Live Preview
- IntelliCode
- Multi-repo git support
- .NET 6!
- Single Project
- Live Visual Tree
- iOS dev on Windows

ANNOUNCING

.NET Conf 2021

November 9 – 11

.NET 6 launch

Free, three-day, virtual event

Save the date!

www.dotnetconf.net



Upcoming Dates

- November 2021
  - .NET 6 GA
  - .NET MAUI Preview
- December 2021 – Q1 2022
  - .NET MAUI Previews monthly
  - .NET MAUI RC
- Q2 2022
  - .NET MAUI GA

November - .NET Conf 2021

	.NET MAUI	.NET MAUI Blazor
Essentials	✅	✅
Platforms		
Android	✅	✅
iOS	✅ Windows ⚠️ Mac	✅ Windows ⚠️ Mac
macOS	⚠️ Mac	⚠️ Mac
Windows	✅	✅
Cross-platform Features		
Controls	⚠️ Preview	⚠️ Preview
Layouts	⚠️ Preview	⚠️ Preview
Productivity		
Single Project	✅	✅
XAML Hot Reload	✅	✅
.NET Hot Reload	✅	✅
Hot Restart for iOS	✅	✅
Tooling		
Visual Studio 2022 Windows	✅	✅
Visual Studio 2022 for Mac	⚠️	⚠️
.NET Upgrade Assistant	✅	n/a
Qualities		
Performance	⚠️	✅
Quality	⚠️	✅

Incomplete:

- Multi-window
- Unpackaged Deploy
- Desktop menus
- Trimming
- Picker (macOS)
- Dual screen lib
- Maps lib
- Fluent for Windows

✅ - GA
⚠️ - Incomplete
🛑 – Blocked/Unavailable

Q2 2022

	.NET MAUI	.NET MAUI Blazor
Essentials	✅	✅
Platforms		
Android	✅	✅
iOS	✅	✅
macOS	✅	✅
Windows	✅	✅
Cross-platform Features		
Controls	✅	✅
Layouts	✅	✅
Productivity		
Single Project	✅	✅
XAML Hot Reload	✅	✅
C# Hot Reload	✅	✅
Hot Restart for iOS	✅	✅
Tooling		
Visual Studio 2022 Windows	✅	✅
Visual Studio 2022 for Mac	✅	✅
.NET Upgrade Assistant	✅	n/a
Qualities		
Performance	✅	✅
Quality	✅	✅

✅ - GA
⚠️ - Incomplete
🛑 – Blocked/Unavailable

Thanks for joining!

David Ortinau | Principal Program Manager | Microsoft

david.ortinau@microsoft.com | @davidortinau

